
An NTA-Based Measure of the Double Buffer Hypothesis

First-Order Approximation for the DFD Simulation Laboratory

1. Purpose and Scope

The Double Buffer Hypothesis proposes that economies have historically absorbed the macroeconomic consequences of demographic transition through two structural mechanisms: public debt expansion (the fiscal buffer) and natural capital utilization (the environmental buffer). As both buffers become increasingly constrained, demographic pressures may become significantly more binding.

Embedding this hypothesis fully within a DSGE-OLG framework is technically demanding and risks producing a model whose outputs are too sensitive to untestable assumptions to be policy-relevant. This note proposes a complementary approach: an accounting-based measure grounded in the National Transfer Accounts (NTA) methodology that can serve as a diagnostic indicator alongside the DFD simulation engine. As an accounting measure it captures no behavioral responses or interaction effects, but as a first-order approximation it is empirically grounded, methodologically transparent, and directly communicable to policy audiences.

2. Starting Point: The NTA Lifecycle Deficit

NTA decomposes economic flows by age. For each age a and period t , the lifecycle deficit is:

$$LD(a,t) = C(a,t) - YL(a,t)$$

where $C(a,t)$ is age-specific consumption and $YL(a,t)$ is age-specific labor income. The aggregate lifecycle deficit is:

$$LD(t) = \int_0^{\omega} LD(a,t) * p(a,t) da$$

where $p(a,t)$ is the population age distribution. The demographic transition operates directly on $p(a,t)$: as fertility falls and populations age, weight shifts toward ages where $LD(a,t) > 0$, increasing aggregate pressure even if individual age profiles remain unchanged. This is the demographic driver of buffer demand.

3. First Buffer Consumption: The Fiscal Gap

From the debt dynamics formalization, the debt-stabilizing total deficit as a share of GDP is:

$$def^*(t) = g_n(t) * b(t)$$

where $g_n(t)$ is the nominal GDP growth rate and $b(t)$ is the debt-to-GDP ratio. The fiscal requirement driven by demographics is approximated by the public sector component of the lifecycle deficit. NTA distinguishes

public transfer inflows and outflows by age:

$$\tau_{g(a,t)} = \tau_{+}(a,t) - \tau_{-}(a,t)$$

where $\tau_{+}(a,t)$ are public transfer inflows and $\tau_{-}(a,t)$ are public transfer outflows by age. The aggregate public transfer deficit is:

$$PTD(t) = \int_0^\omega \tau_{g(a,t)} * p(a,t) da$$

The first buffer consumption rate is then:

$$B_1(t) = PTD(t)/Y(t) - def^*(t)$$

When $B_1(t) > 0$, the demographic structure is generating public transfer demands that exceed what is fiscally sustainable at current debt levels and growth rates — the buffer is being consumed. When $B_1(t) < 0$, fiscal space remains available. For Mexico, with an RFSP of approximately 5% of GDP against a stabilizing deficit of 3.0-3.5% of GDP (at $b = 50\%$ and nominal growth of 6-7%), $B_1(t)$ is already clearly positive.

Note on fiscal dominance: when $B_1(t)$ is persistently positive and growing, governments face incentives to tolerate higher inflation to increase g_n artificially, reducing the real debt burden. This is a third-order deterioration — it erodes real savings and pension assets precisely when an aging population most depends on them. The model should be capable of producing fiscal dominance endogenously as a scenario outcome.

4. Second Buffer Consumption: The Environmental Gap

The environmental buffer enters as net natural capital depletion. The environment is modeled as a stock of natural capital N entering the production function:

$$Y = F(K, L, N)$$

Its dynamics follow:

$$N_{t+1} = N_t - E_t + R_t$$

where E_t is extraction or emissions and R_t is natural regeneration. Sustainable use requires $E_t \leq R_t$. When extraction exceeds regeneration, the economy is consuming natural capital exactly as the government consumes fiscal space through a deficit exceeding def^* .

The most comprehensive cross-country measure available is the World Bank's adjusted net savings framework, which computes net natural capital depletion as:

$$D_N(t) = \text{energy depletion} + \text{mineral depletion} + \text{net forest depletion} + \text{CO2 damage}$$

All terms are expressed as shares of GNI. The second buffer consumption rate is:

$$B_2(t) = D_N(t) / Y(t)$$

When $B_2(t) > 0$ — which holds for virtually every middle-income economy — natural capital is being drawn down to sustain current output. The structural parallel with the first buffer is exact: production today against environmental costs tomorrow, just as deficit spending today against tax obligations tomorrow.

5. The Composite Double Buffer Indicator

The total buffer consumption rate is:

$$B(t) = B_1(t) + B_2(t)$$

This is the first-order approximation of the rate at which the economy is shifting costs to the future across both dimensions simultaneously. It is an accounting identity — transparent, decomposable, and directly interpretable as a percentage of GDP.

The demographic contribution to $B_1(t)$ can be isolated by projecting $PTD(t)$ forward under alternative population scenarios while holding age profiles and fiscal parameters constant. The difference between baseline and stress scenario projections isolates the purely demographic component — a clean counterfactual that NTA is exceptionally well suited to support.

6. Projection Structure for DFD

The measure would be computed at three temporal moments, consistent with DFD scenario discipline:

Horizon	Description	Role
$B(t_0)$	Current observed buffer consumption, calibrated to data	Diagnostic baseline
$B(t_0 + 20)$	Medium-term projection under baseline demographic scenario	Primary policy-relevant horizon; demographic momentum largely determined
$B(t_0 + 40)$	Long-run projection under alternative scenarios	Presented as range, not point estimate; uncertainty is first-order

7. Data Requirements and Country Coverage

For $B_1(t)$: NTA age profiles for consumption, labor income, and public transfers are required. These are available for several Latin American economies — Mexico, Brazil, Chile, Colombia, Costa Rica — through the NTA project database. Population age structures come from CELADE and CONAPO (primary sources), cross-checked against UN WPP with the data quality protocol recommended following the Fernandez-Villaverde critique of WPP systematic bias. Fiscal parameters (b , g_n , actual deficit) come from national sources and IMF Article IV assessments.

For $B_2(t)$: World Bank adjusted net savings data provides $D_N(t)$ for all DFD sample countries on a consistent basis. Methodological limitations include: CO2 damage estimates are sensitive to the social cost of carbon assumed; forest depletion accounting varies by country; and subsoil asset valuation is imprecise for countries with heterogeneous resource endowments. These limitations should be documented and sensitivity analysis conducted.

8. Role Within the DFD Architecture

$B(t)$ provides three things that the DSGE-OLG model cannot easily supply. First, direct demographic attribution: changes in $B_1(t)$ can be decomposed into contributions from age structure shifts, age profile

changes, and fiscal parameter changes — which the OLG model conflates through behavioral responses. Second, cross-country comparability: the NTA methodology is standardized, so $B(t)$ can be computed consistently across Latin American economies. Third, communicability: $B(t)$ expressed as a percentage of GDP is immediately interpretable for policy audiences.

The limitation is clear and must be stated explicitly in any publication: $B(t)$ contains no behavioral content, no interaction effects, and no endogenous policy response. It establishes the magnitude of the problem — the rate at which costs are being shifted to the future — while the DFD simulation engine addresses dynamics and policy responses. The two are complementary, not competing analytical tools.

9. Immediate Next Steps

Three tasks follow from this specification. First, compile NTA age profiles for the DFD sample countries and verify consistency with current population data under the triangulation protocol (vital registries vs. WPP). Second, extract adjusted net savings series from World Bank for 2000-2024 for all sample countries and document data quality issues by country. Third, compute $B_1(t_0)$ and $B_2(t_0)$ for Mexico as a pilot case, using the fiscal parameters from the debt buffer presentation as inputs, and present results to the Core Team for validation before extending to the full sample.